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An Interpretable Framework for Stock Market Forecasting using Long Short-Term Memory (LSTM) Networks with SHAP - Explainable AI (XAI) Method

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Abstract: Predicting stock market movements is a challenging and ever-changing process because it is by a variety of unpredictable elements. In this study, we explore the use of deep learning, specifically Long Short-Term Memory (LSTM) networks, to estimate the upcoming stock prices of HDFC Bank. The training process of the model utilizes past stock closing prices to learn patterns and make future predictions from 2020 to 2025, sourced from Yahoo Finance. The time series data is preprocessed using normalization techniques, and sequences of past observations are structured to train the LSTM architecture effectively. To make the model's predictions easier to understand and explain, Explainable AI methods, particularly SHAP (SHapley Additive exPlanations), are employed. The proposed model demonstrates strong predictive performance, achieving a Root Mean Squared Error (RMSE) of 0.0246, indicating that the average difference between the predicted and actual values is quite low. Additionally, the R^2 (coefficient of determination) score was 0.9489, suggesting that approximately 94.89% of the variability in the actual values is explained by the model's predictions. These results highlight the model's capability to accurately capture trends in stock price movement while also providing valuable insights into feature contributions. Unlike traditional statistical models, LSTM networks can retain long-term dependencies, enabling them to detect time-based patterns effectively. The integration of explainability not only adds transparency to the predictions but also enhances the trustworthiness of the model in real-world applications. Furthermore, this approach can be adapted to other financial instruments or extended with external indicators for broader market analysis. The combination of LSTM and SHAP enables a robust, interpretable framework for stock market forecasting that supports data-driven investment strategies.

Keywords: Long Short-Term Memory, Root Mean Squared Error, Explainable AI, Quality,

I. INTRODUCTION

Forecasting the behavior of the stock market has consistently posed difficulties due to its complex and unpredictable nature due to the market's inherently volatile and non-linear behavior [1]. Accurate forecasting of stock prices can provide significant advantages to investors, portfolio managers, and financial analysts by aiding in risk management and investment decision-making [2]. Traditional models, including statistical and econometric approaches, they frequently struggle to recognize the complex patterns and time-based relationships that are characteristic of stock price movements, limiting their effectiveness in dynamic market environments [3].

Recent developments in deep learning have opened new avenues for tackling time-series prediction problems. Long Short-Term Memory (LSTM) networks are a unique kind of recurrent neural network (RNN) that are widely used because they can effectively handle and learn from sequences of data over time with long-term dependencies [4]. Unlike traditional models, LSTM networks are capable of retaining crucial information for long durations, allowing them to handle sequences with long-term dependencies more effectively, making them well-suited for stock market data where historical trends and seasonality often influence future prices. This capability enables the LSTM to outperform many traditional forecasting methods in terms of accuracy and robustness [5].

This research combines the strengths of LSTM networks and SHAP explainability to develop a reliable and interpretable stock price prediction model for HDFC Bank [6]. Using historical market data from 2020 to 2025, the LSTM model is trained to capture temporal dependencies,

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while SHAP is employed to interpret the way the model arrives at its decisions is carefully analyzed. To assess how well the model performs, it is thoroughly tested using evaluation measures like Root Mean Square Error (RMSE) and R^2, demonstrating strong predictive capabilities. By integrating explainability, this approach not only enhances forecasting accuracy but also fosters trust and transparency, making it valuable for practical financial analysis and investment strategies [7].

Furthermore, the integration of explainable AI with deep learning models addresses growing concerns regarding model accountability and ethical use of AI in finance [8]. As automated trading and algorithmic investment strategies become more prevalent, stakeholders require assurance that model predictions are not only accurate but also understandable and justifiable. This transparency is essential for regulatory compliance, risk assessment, and building confidence among users [9]. Moreover, the techniques developed in this study have the potential to be extended beyond stock prices to other financial instruments such as commodities, currencies, and derivatives, broadening their practical application. The fusion of LSTM and SHAP presents a promising step toward more interpretable, trustworthy, and effective financial forecasting tools in an increasingly data-driven market environment [10].

II. LITERATURE SURVEY

Forecasting stock market prices has become a major area of research because it plays a vital role in guiding investment and financial planning. Initially, most prediction methods were based on statistical models such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) to analyze time-series data. These models assume linear relationships and often fail to capture the complex, non-linear behaviors typical of stock market data [11].

The introduction of Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, marked a significant advancement in financial Time-series forecasting means predicting future data based on past values. LSTM models are good at this because they can remember important information for a long time using a memory cell that manages what to keep or forget, making them highly effective for sequential Data like stock prices often change over time [12]. Research has shown that LSTM models usually give better results than older forecasting methods when working with this kind of data by better capturing trends and patterns over time. For example, [13] showcased that LSTM networks could successfully forecast stock price movements by learning from historical data, outperforming conventional machine learning models. However, despite these improvements, LSTM models often lack transparency, limiting their interpretability and acceptance in critical financial applications.

Several hybrid approaches have emerged that combine the strengths of LSTM models with explainability frameworks. For instance, recent studies have utilized LSTM models to predict stock prices while employing SHAP values to analyze the influence of each time-step and feature. These approaches enable practitioners to not only achieve high prediction accuracy but also gain insights into which temporal features drive market behavior [14].

Despite these advancements, challenges remain in effectively handling the noisy, volatile nature of stock market data and incorporating external factors such as news sentiment, macroeconomic indicators, and geopolitical events into predictive models. Future research is focusing on multimodal data fusion, combining textual news data and numerical time series, along with more advanced explainability techniques that can handle high-dimensional inputs. The integration of attention mechanisms within LSTM models is also being explored to enhance both prediction performance and interpretability. This evolving landscape indicates that combining deep learning with explainable AI holds great promise for developing robust, trustworthy, and actionable stock market forecasting tools [15]. Older statistical methods like ARIMA and GARCH have been commonly used in the past to predict stock market trends due to their simplicity in modeling linear time series data. However, these models They usually struggle to understand the complex and nonlinear trends, as well as long-term relationships, that exist in financial market data [16]. Machine learning algorithms like Stock price prediction has been done using machine learning models like Support Vector Machines (SVM) and Random Forests, which help in analyzing patterns in financial data with improved accuracy over classical methods. These models benefit from feature engineering but generally lack the ability to model temporal sequences effectively, which limits their predictive power in stock market applications [17]. Recurrent Neural Networks (RNNs), and especially LSTM networks, are known for performing well when working with data that comes in a sequence, like time-based or ordered information by capturing long-term dependencies and nonlinear trends. Many studies report that LSTM-based models outperform conventional machine learning techniques in financial forecasting tasks [18]. Despite their performance advantages, LSTM models are often criticized for being "black-box" models that lack interpretability. To address this, explainable AI methods such as SHAP (SHapley Additive exPlanations) have been introduced, enabling detailed insights into feature contributions, and improving the trustworthiness of deep learning models in finance [19]. Recent hybrid approaches combining LSTM with explainability frameworks like SHAP have shown promise in providing accurate predictions while also offering interpretability. These methods help analysts understand which features or past time steps most significantly impact the model's predictions, facilitating more informed investment decisions [20].

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III. ALGORITHMS USED

1. Understanding Long Short-Term Memory (LSTM) Networks

This project uses a Long Short-Term Memory (LSTM) network as the main tool to forecast upcoming stock prices by analyzing past market data. Stock market data is inherently sequential and time-dependent, where past prices influence future movements. Traditional neural networks struggle to capture such temporal relationships effectively, especially over long time periods. LSTM networks address this challenge They include memory cells and special gates that help the system decide which information to keep and which to remove as time passes.

By feeding the model with sequences of past closing prices, the LSTM learns Stock market data contains various patterns and trends that can be analyzed. The LSTM model uses three gates—forget, input, and output—to decide which information to ignore, which to add, and what to pass on, helping it pay attention to the most important details important historical data while ignoring irrelevant fluctuations. This makes LSTM highly suitable for modeling the nonlinear and noisy behavior of stock prices. In this project, a stacked LSTM architecture is employed to capture complex temporal dependencies at multiple levels, improving prediction accuracy. The ability of LSTM to remember long-term dependencies helps the model anticipate price changes better than simpler methods, thus enhancing the quality of stock market forecasts.

LSTM Network Structure:

The LSTM model designed for this project is a **stacked LSTM network** The model is built with three LSTM layers stacked one after another, and then a Dense layer at the end to produce the result. This arrangement helps the model understand and learn complicated patterns in the data patterns in sequential stock price data more effectively.

i. **First LSTM Layer:** This layer consists of 50 memory units (neurons) and is set up to produce an output in the form of a sequence, rather than a single value hidden states for each time step. This allows the next LSTM layer to receive a full sequence as input rather than just the final output.
ii. **Second LSTM Layer:** Similar to the first, it also contains 50 units and returns sequences, allowing for deeper temporal feature extraction.
iii. **Third LSTM Layer:** This layer has 50 units but returns only the final hidden state, summarizing the sequence information learned by the previous layers into a fixed-length vector.
iv. **Dense Layer:** Following the LSTM layers, a fully connected layer containing a single neuron is included to output the final stock price prediction.

The input shape for the model is (100, 1), where 100 represents the number of previous time steps (days) used to predict the next value, and 1 represents the single feature — the normalized closing price. This architecture This helps the model understand and learn patterns from both recent data and information from far back in the sequence. stock price data, improving forecasting performance.

2.Min-Max Scaling

To normalize the dataset within the range [0, 1], each feature value x is scaled as

x_{min} and x_{max} are the minimum and maximum values of the feature in the dataset,

x' is the scaled value.

This ensures that all features are on the same scale, which helps in faster and more stable training of neural networks.

3. Mean Squared Error (MSE) Loss Function

Min-Max scaling is a method used to adjust data values so they fall within a specific range, usually from 0 to 1. This is done by taking each value, subtracting the smallest value in the dataset, and then dividing by the difference between the largest and smallest values. As a result, all the data points are resized while keeping their original relationships. This technique is helpful in machine learning, as it puts all features on the same scale and helps algorithms perform better.

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4. Adam Optimizer: An Efficient Method for Fast and Accurate Model Training

Adam (Adaptive Moment Estimation) is a powerful method used to improve how deep learning models learn. It mixes the strengths of two other optimizers: AdaGrad, which is helpful when the data has many zero values, and RMSProp, which performs well when the data changes over time. Adam maintains adaptive learning rates for each parameter by computing individual estimates of both the first-order moment (mean) and the second-order moment (uncentered variance) of the gradients.

5. SHAP (SHapley Additive exPlanations)

SHAP helps us understand a machine learning model's prediction by showing how each feature contributes to the final result. It uses a method from game theory called Shapley values to fairly divide the prediction among all input features. So, the output for a particular input can be seen as the sum of the effects of each feature.:

$$f(x)= \phi_0+ \sum \phi_i$$

where

- ϕ_0 is the expected output over the training data (baseline),
- ϕ_i shows how much feature ***i*** adds to the final prediction in the SHAP method. It tells us the importance or effect of that feature.
- M represents the total count of input variables or features used in the model.

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The SHAP value (ϕi) for a feature i is calculated by averaging its contribution to the prediction across all different combinations of features, with each contribution weighted appropriately. Fig. 1 shows the flow diagram of SHAP based stock market forecasting.

IV. METHODOLOGY

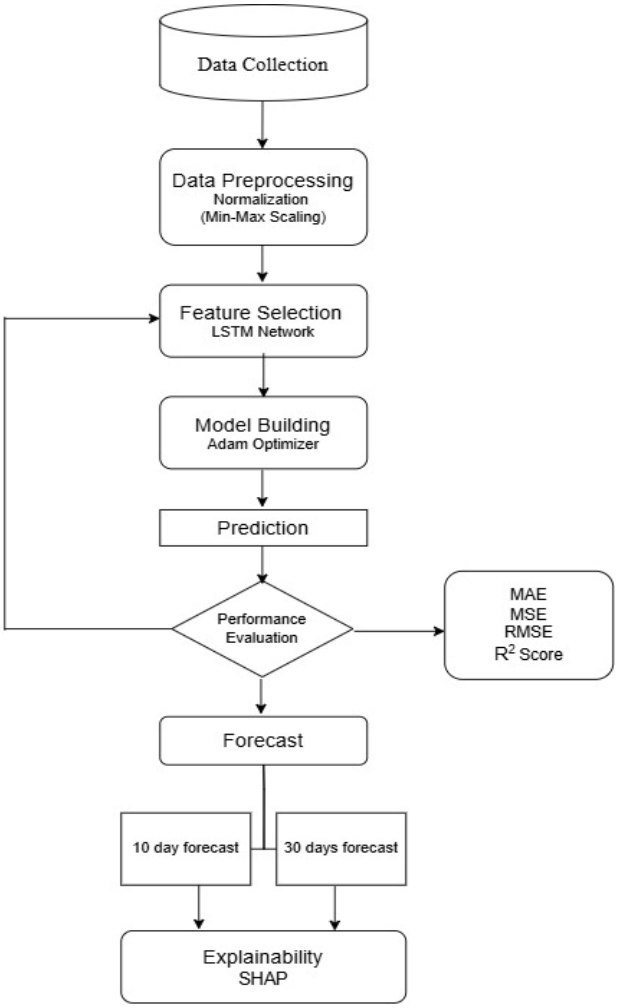


fig 1. Flow diagram

1) **Data Acquisition:**
The historical stock data of HDFC Bank (ticker: HDFCBANK.NS) was collected using the yfinance Python library, covering the period from January 1, 2020, to May 2, 2025. The dataset includes daily stock features such as Open, High, Low, Close, Adj Close, and Volume. The study specifically focuses on the 'Close' price, which is typically used for market trend analysis.

2) Data Preprocessing:
The preprocessing phase includes the following steps:

I. The data is processed to remove any unwanted or incorrect information, and the focus is placed on the 'Close' price values for further analysis.
II. The data values are scaled down so that they fit within the range of 0 to 1 using MinMaxScaler to enhance model convergence.
III. The data is divided so that 65% is used for training the model, while the remaining 35% is reserved for testing its performance.
IV. A sliding window method is applied to generate sequences of time series data by moving a fixed-size window over the dataset sequences with look-back period of 100 days, where each sequence predicts the next day's closing price.

3) Model Architecture

A stacked LSTM neural network is implemented using the Keras API with TensorFlow backend. The architecture:

- The model consists of three LSTM layers, each containing 50 units.
- A fully connected (dense) layer is used at the end to make continuous value predictions in regression tasks.
- The model is built using the Mean Squared Error as the loss function to measure prediction accuracy, and it uses the Adam optimizer to improve learning during training.

The model is trained over 100 cycles (epochs), with each training step processing 64 samples at a time (batch size) test set for validation during training.

4) Predictions

The graph shown in fig 2 illustrates A comparison is made between the real values and the values predicted by the model to check how accurate the prediction is the HDFC Bank stock using an LSTM model. As seen in the plot, the orange line, which shows the predicted results, stays very close to the blue line representing the real values. This means the model is able to understand and follow the actual data pattern well underlying trend and fluctuations of the stock prices with good accuracy.

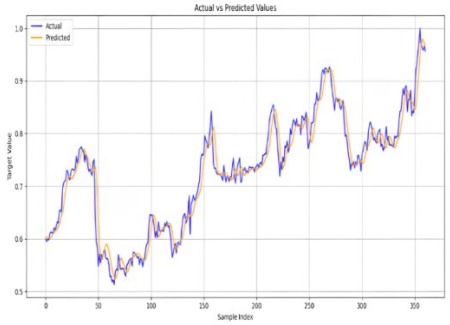


fig 2. Actual vs Predicted Graph

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5) Forecasting and Evaluation

Post training, the model's performance is evaluated using key metrics:

- **Mean Absolute Error (MAE)** was **0.01796**, showing the average difference between predicted and actual values.
- **Mean Squared Error (MSE)** came out to **0.000604**, indicating how much the prediction errors were spread out.
- **Root Mean Squared Error (RMSE)** was **0.02458**, providing an error measure in the same unit as the data.
- **R^2 Score** was **0.9489**, meaning the model explained about **94.89%** of the variation in the target values.

1. Forecast for next 10 days:

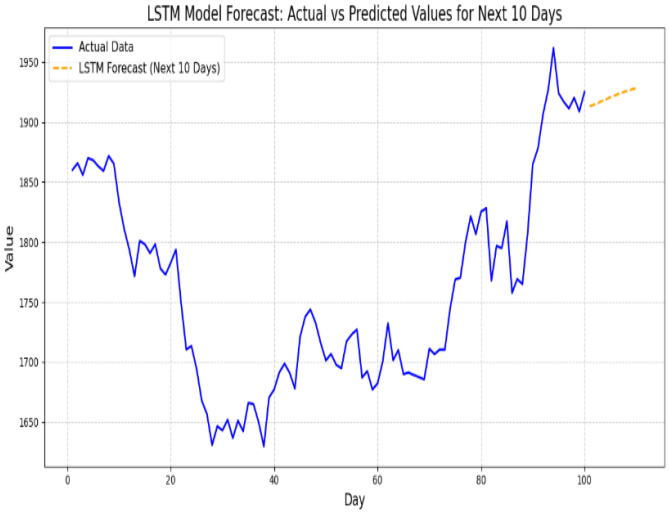


fig 3. Forecast for next 10 days

```
Predicted Stock Prices for Next 10 Days:
Day 1: 1908.51
Day 2: 1910.42
Day 3: 1911.50
Day 4: 1912.77
Day 5: 1914.48
Day 6: 1916.52
Day 7: 1918.72
Day 8: 1920.90
Day 9: 1922.97
Day 10: 1924.85
```

fig 4. Predicted stock price for next 10 days

2. Forecast for next 30 days:

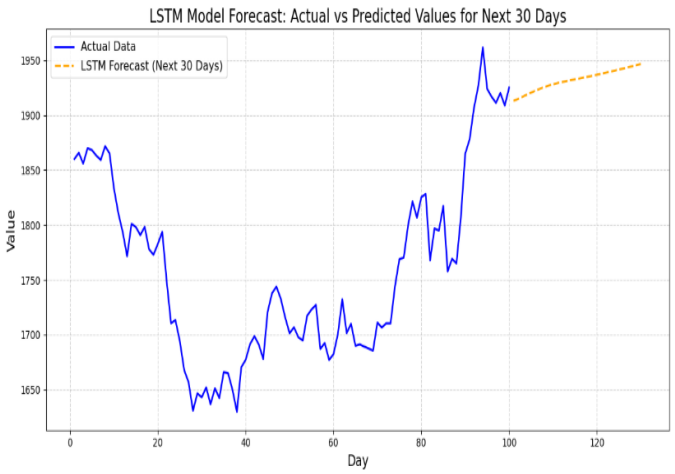


fig 5. Forecast for next 30 days

```
Predicted Stock Prices for Next 30 Days:
Day 1: 1908.51
Day 2: 1910.42
Day 3: 1911.50
Day 4: 1912.77
Day 5: 1914.48
Day 6: 1916.52
Day 7: 1918.72
Day 8: 1920.90
Day 9: 1922.97
Day 10: 1924.85
Day 11: 1926.52
Day 12: 1928.00
Day 13: 1929.30
Day 14: 1930.45
Day 15: 1931.50
Day 16: 1932.48
Day 17: 1933.41
Day 18: 1934.33
Day 19: 1935.25
Day 20: 1936.18
Day 21: 1937.11
Day 22: 1938.05
Day 23: 1939.01
Day 24: 1939.97
Day 25: 1940.93
Day 26: 1941.88
Day 27: 1942.83
Day 28: 1943.77
Day 29: 1944.70
Day 30: 1945.62
```

fig 6. Predicted stock price for next 30 days

6) Explainable AI with SHAP

To interpret the LSTM model's predictions, SHAP (SHapley Additive exPlanations) is used. The SHAP framework explains the contribution of each time step in the input sequence to the predicted output as shown in fig 7,8,9.

- SHAP values are calculated for selected test samples.
- Visualizations include SHAP waterfall plots and summary bar charts.
- The XAI component helps identify which past days significantly influence the prediction, improving model transparency.

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Waterfall plot:

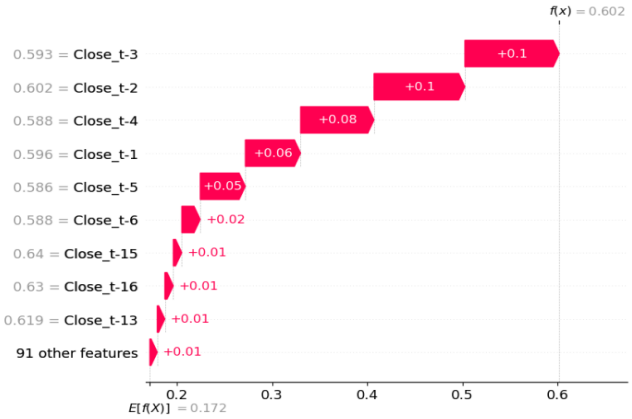


fig 7. Waterfall plot of feature importance

Bar plot:

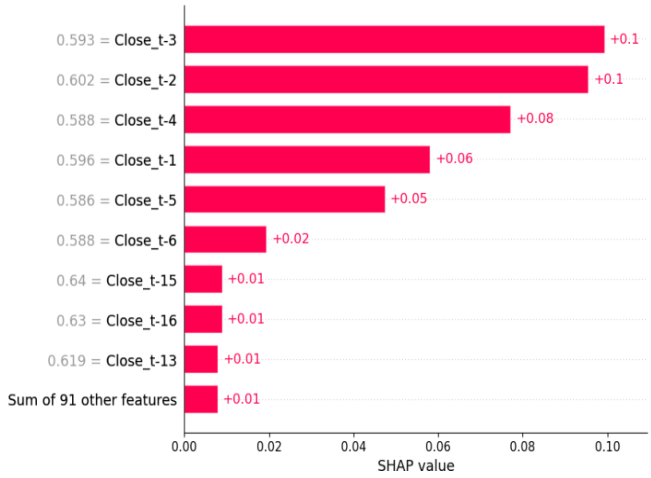


fig 8. Bar plot of feature importance

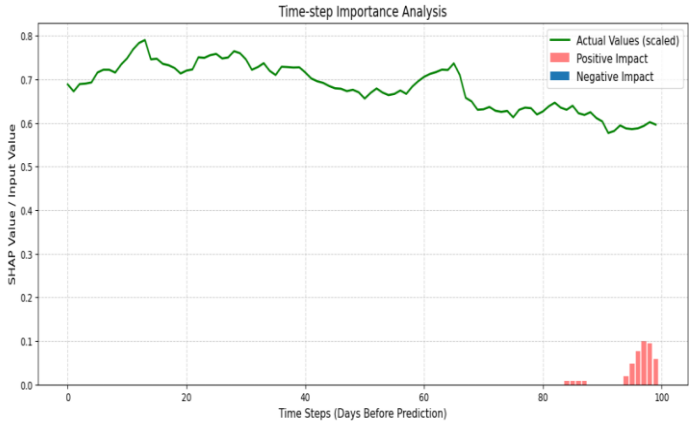


fig 9. Time step importance analysis

V. *RESULT AND ANALYSIS*

Model Performance Metrics

The small MAE and RMSE values show that the model's predicted closing prices are very close to the real ones. The R^2 score of approximately 0.95 further demonstrates that the model can understand and describe a large part of the changes or patterns found in the data.

Visualization of Predictions

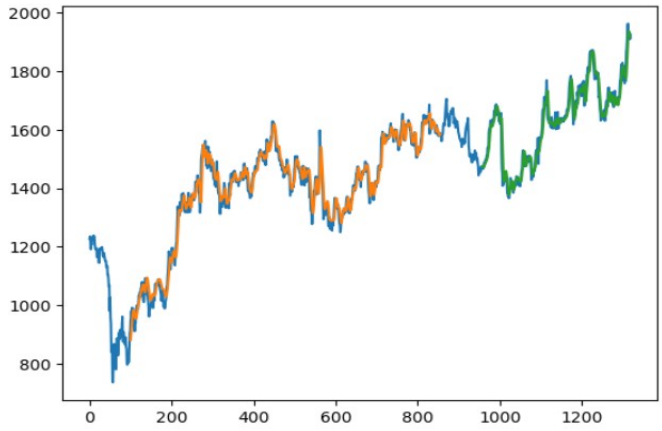


fig 10. Visualization of Prediction

Visualizing the actual values alongside the predicted results helps to better understand how accurately the model is performing as shown in fig 2 and fig 10.

Using the final 100 days of input data, the model was tasked with predicting the next 30 closing prices. The forecasted values were inverse-transformed to obtain real-world price estimations. The predictions were plotted alongside the recent historical data to visualize the continuation of trends. To enhance model transparency and interpretability, SHAP (SHapley Additive exPlanations) was applied to the test predictions. The SHAP framework allowed for identifying which days in the 100-day input window had the most significant influence on the predicted output. Key findings from SHAP analysis: Recent days generally had higher SHAP values, as shown in fig 7, indicating a greater impact on future price prediction.

SHAP waterfall plots as shown in fig 7 showed both positive and negative contributions of specific days, offering an easy-to-understand visual representation that shows the process the model used to make each prediction. The results indicate that LSTM is a powerful model for time series forecasting, especially in financial applications like stock price prediction. This allows stakeholders to understand the model's reasoning, improving trust and usability in real-world decision-making.

VI. *CONCLUSION*

In this research, we built a deep learning model based on Long Short-Term Memory (LSTM) networks to predict stock prices of HDFC Bank, combined with Explainable AI (XAI) techniques to enhance model transparency. The LSTM model demonstrated strong predictive performance, The model performs very well,

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reaching a high accuracy with an R^2 score of 0.9489 and a low Root Mean Squared Error (RMSE) of 0.02458. This shows that it is effective in understanding and learning the patterns over time in the financial data., the integration of SHAP (SHapley Additive exPlanations) provided valuable interpretability by highlighting the influence of specific past days Based on what the model forecasts. Overall, the combination of LSTM's sequence learning capabilities with SHAP-based interpretability presents a powerful framework for stock market prediction. This approach not only delivers accurate forecasts but also addresses the critical need for transparency in AI-driven financial models. SHAP delivered auditable feature importance, which is essential for financial AI regulatory compliance. Unveiled the decision-making of the LSTM consistent with technical analysis rules, enhancing stakeholders' trust. This study can be extended in several ways. Future models may include additional inputs like trading volume, news sentiment, or macroeconomic indicators to improve accuracy. Exploring hybrid models or advanced architectures such as Transformers could further enhance predictions. Real-time deployment and multi-stock forecasting are also promising directions. Lastly, integrating more explainability tools beyond SHAP may offer deeper insights into model decisions.

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